Databases in Biology Research

Data Exploration and Visualisation
2025/2026/2

Anikó Mentes, PhD

2026-04-15

Table of contents

01 **Biological Fundamentals**
DNA, mutations, and why they become data problems

02 **Viral Genomic Databases**
GISAID, NCBI Virus, COVID-19 Data Portal

03 **VCF Files in Viral Genomics**
Format, quality metrics, within-host allele frequency

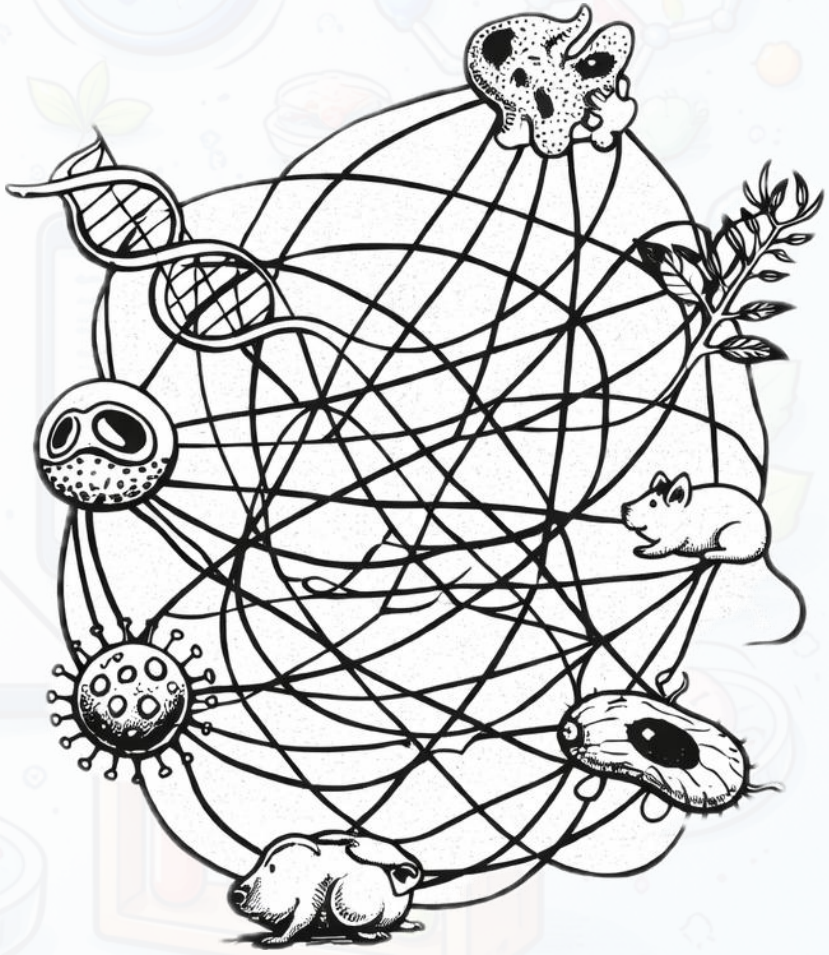
04 **Group Studies**
CoVEO (SARS-CoV-2) · SewageGPT (wastewater metagenomics)

05 **AI-Assisted Analysis**
GitHub Copilot, VSCode, MCP servers, and Text-to-SQL

01

Biological Fundamentals

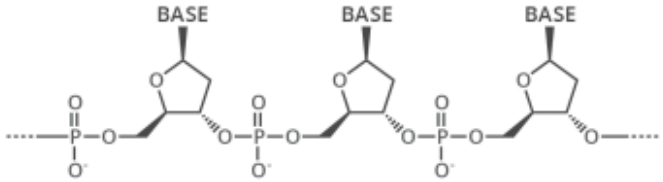
From molecules to data structures



The central dogma

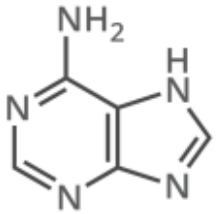
Why is DNA such a powerful data carrier?

The sugar-phosphate backbone

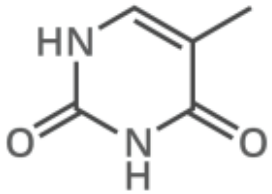


DNA is a polymer made up of units called nucleotides. The nucleotides are made of three different components: a sugar group, a phosphate group, and a base. There are four different bases: adenine, thymine, guanine and cytosine.

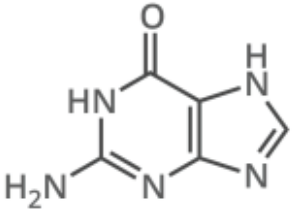
A Adenine



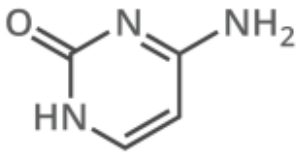
T Thymine



G Guanine

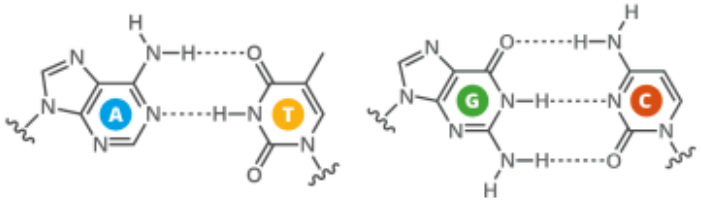


C Cytosine



What holds DNA strands together?

DNA strands are held together by hydrogen bonds between bases on adjacent strands. Adenine (A) always pairs with thymine (T), while guanine (G) always pairs with cytosine (C). Adenine pairs with uracil (U) in RNA.



From DNA to proteins

The bases on a single strand of DNA act as a code. The letters form three letter codons, which code for amino acids - the building blocks of proteins.

An enzyme, RNA polymerase, transcribes DNA into mRNA (messenger ribonucleic acid). It splits apart the two strands that form the double helix, then reads a strand and copies the sequence of nucleotides. The only difference between the RNA and the original DNA is that in the place of thymine (T), another base with a similar structure is used: uracil (U).

In multicellular organisms, the mRNA carries genetic code out of the cell nucleus, to the cytoplasm. Here, protein synthesis takes place. 'Translation' is the process of turning the mRNA's 'code' into proteins. Molecules called ribosomes carry out this process, building up proteins from the amino acids coded for.

https://www.compoundchem.com/2015/03/24/dna/

DNA as biological data

Digital properties

A biological sequence is fundamentally a digital string

ATGGCCTACAGCTGAGCTGCGTACGCAGC

4 letters only

A, T, G, C. Encodes 2 bits of information each. Like a base-4 number system.

Massive but compressible

Human genome: ~725 MB as plain text. SARS-CoV-2: ~30 KB. Highly repetitive → compresses well.

Queryable

Databases let us search billions of characters for specific patterns, variants, and positions.

Meaningful structure

The sequence isn't random.
Genes, codons, regulatory regions all have specific biological roles.

Source: https://www.compoundchem.com/2015/03/24/dna/

From sample to mutation record

How do we get DNA information?

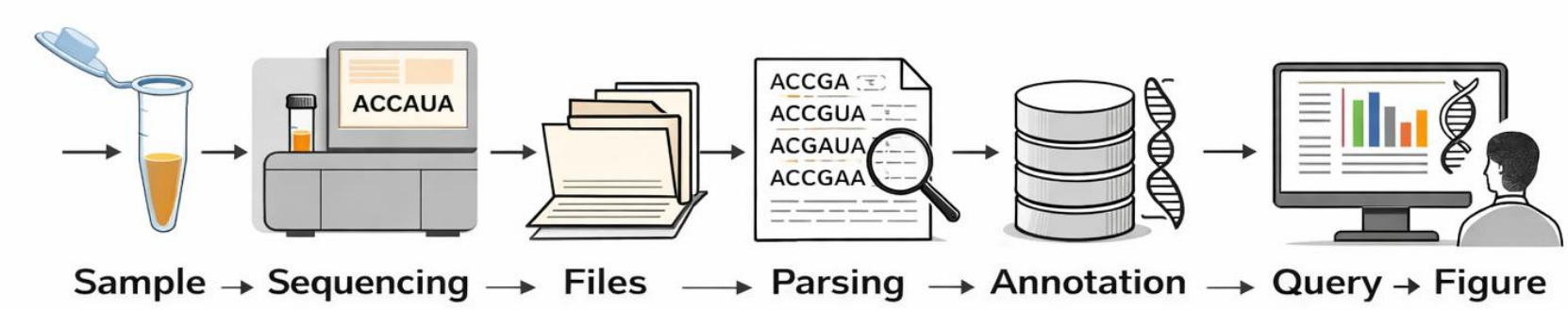
- we collect samples and sequence the genetic material

What is a mutation?

- a change at a specific genomic position

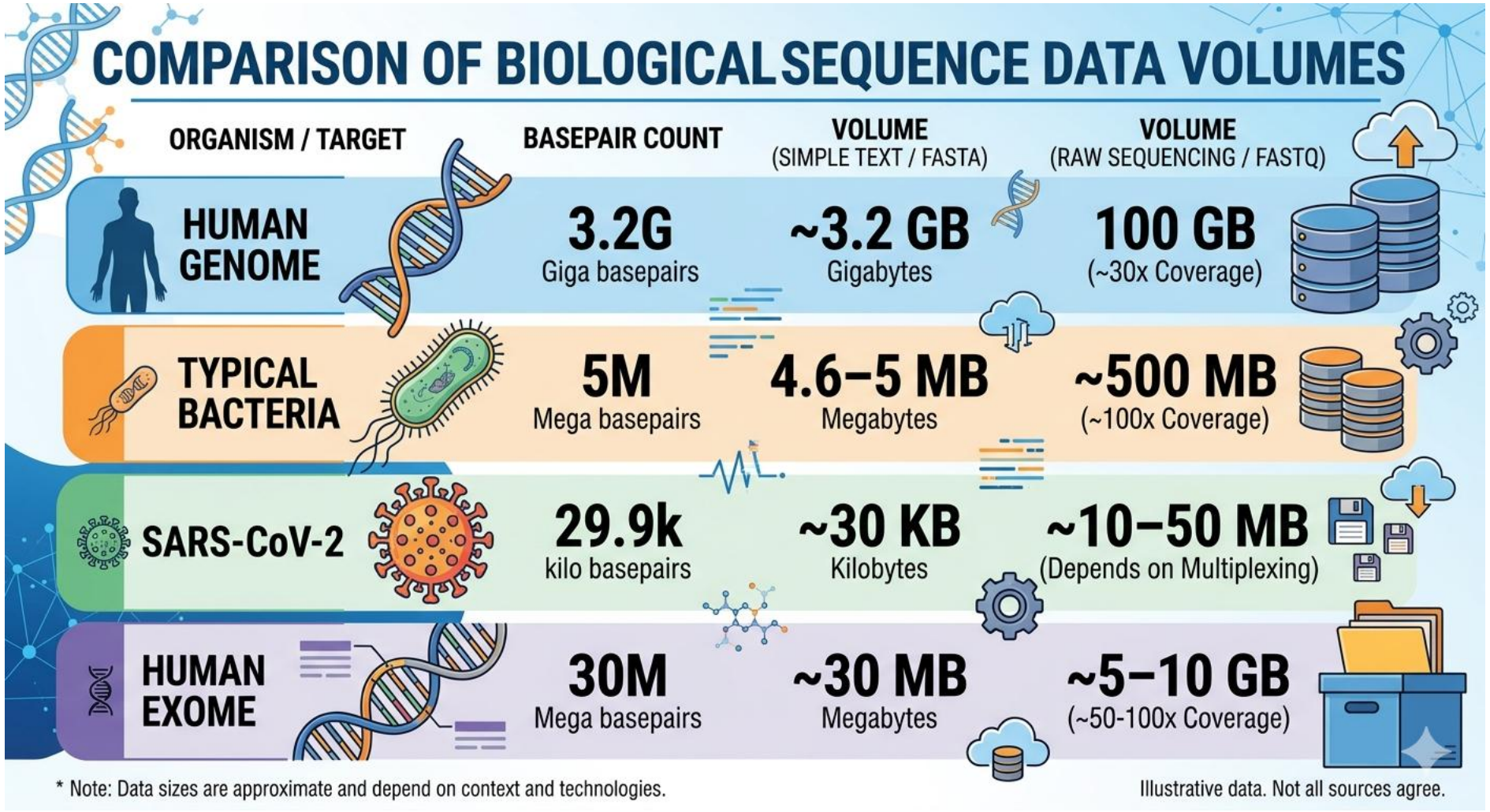
Why does this matter in viral genomics?

- mutations help us track viral evolution, variants, and diagnostic relevance



Source: https://www.compoundchem.com/2015/03/24/dna/

Biological sequence data volumes



A small genome can still produce large raw sequencing datasets. Sequence size is not the same as sequencing data size.

Mutation as database record

POSITION:
23063 G → C
(N501Y in Spike protein)

Reference: GCCTACATGAGCTGTACAGC
Alternative: GCCTACATCAGCTGTACAGC

SNP

This becomes a database row:

Field	Value
POS	23063
REF	G
ALT	C
AF	0.997
DP	2341
EFFECT	N501Y (missense)
GENE	S (Spike)

The scale problem

Example – SARS-CoV-2 sequencing data

GISAID database

18M+ samples

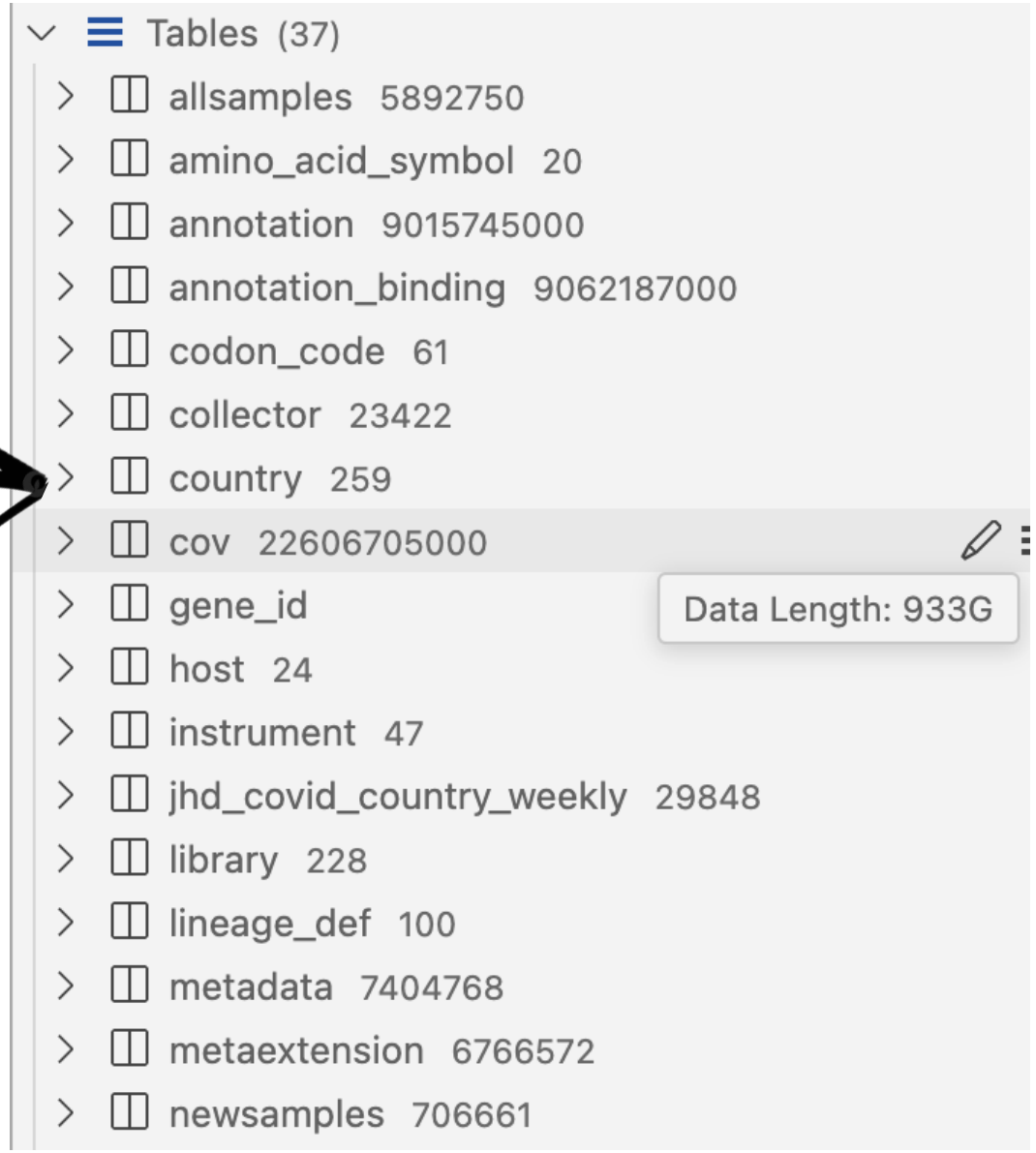
CoVEO database

7,4M samples, 3,7B+ mutation records

Average 500 mutation records / sample

+Processed pipelines

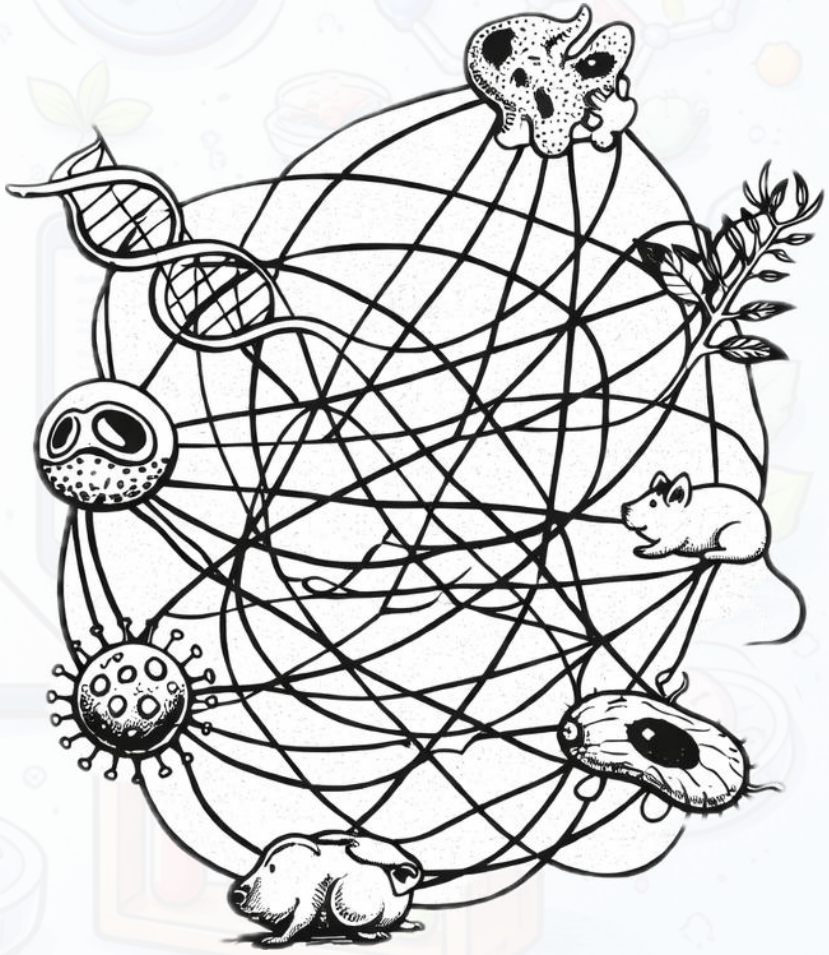
More than 3,7 TB of data



02

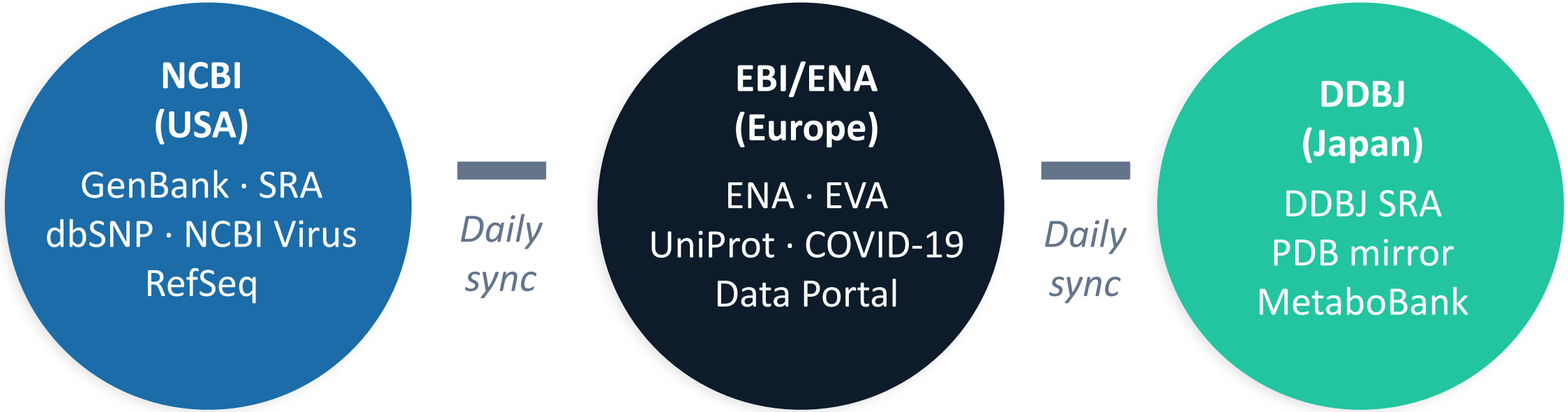
Viral Genomic Databases

GISAID · NCBI Virus · COVID-19 Data Portal



The global data infrastructure

INSDC: one submission, three mirrors, open forever



Pandemic-era specialised databases (on top of INSDC foundation)

GISAID EpiCoV · COVID-19 Data Portal · Nextstrain · CoV-Spectrum · outbreak.info

Key databases during COVID- 19 pandemic

GISAID EpiCoV

gisaid.org

The primary repository for SARS-CoV-2 genomes. >16 million sequences. Requires registration. Stores consensus FASTA — not VCF. Geotemporal metadata + Pangolin lineage.

COVID-19 Data Portal

covid19dataportal.org

EBI-hosted open-access portal. No registration required. Raw FASTQ + assemblies downloadable. ~2.8M samples processed in our pipeline.

NCBI Virus

ncbi.nlm.nih.gov/labs/virus

Curated viral genomes from GenBank + RefSeq. Contains NC_045512.2 — the canonical SARS-CoV-2 reference. Searchable by taxon, host, country, date.

RefSeq Viral

ncbi.nlm.nih.gov/refseq

NCBI-curated, non-redundant reference sequences. Always use these — not raw GenBank — as alignment targets. NC_045512.2 is here.

Nextstrain

nextstrain.org

Real-time phylogenetic trees with mutation annotations. Shows variant emergence timeline. Uses a curated subset of high-quality genomes.

CoV-Spectrum

covspectrum.org

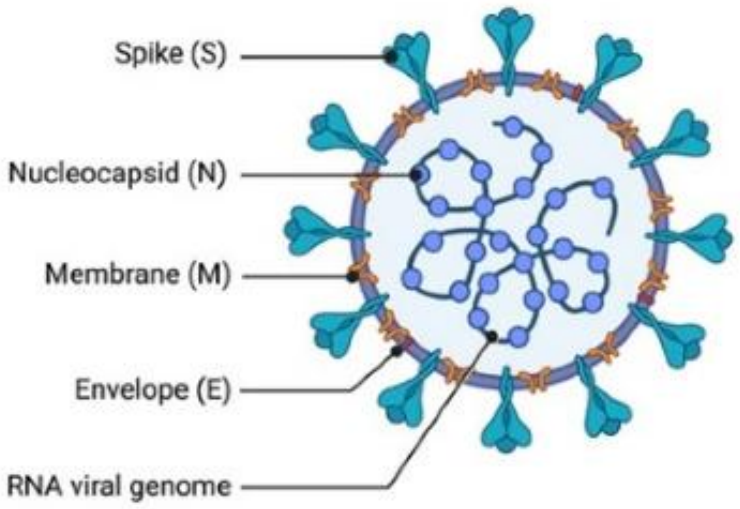
Variant frequency over time per country. Integrates GISAID data. Excellent for epidemiological trend queries without SQL.

Human vs viral genomics

Feature	Human Genome	SARS-CoV-2 Genome
Size	3.2 billion bp	29,903 bp
Type	Double-stranded DNA	Single-stranded RNA
Ploidy	Diploid (2 copies)	Haploid (1 copy)
Mutation rate	~1.2×10^{-8} per bp/gen	~1.5×10^{-6} per bp/replication
Pop. size	8 billion humans	10^{10} particles per host
Reference	GRCh38 (stable)	NC_045512.2 (drifts over time)
Key DB	gnomAD, ClinVar	GISAID, CoVEO, NCBI Virus

SARS-CoV-2 genome structure

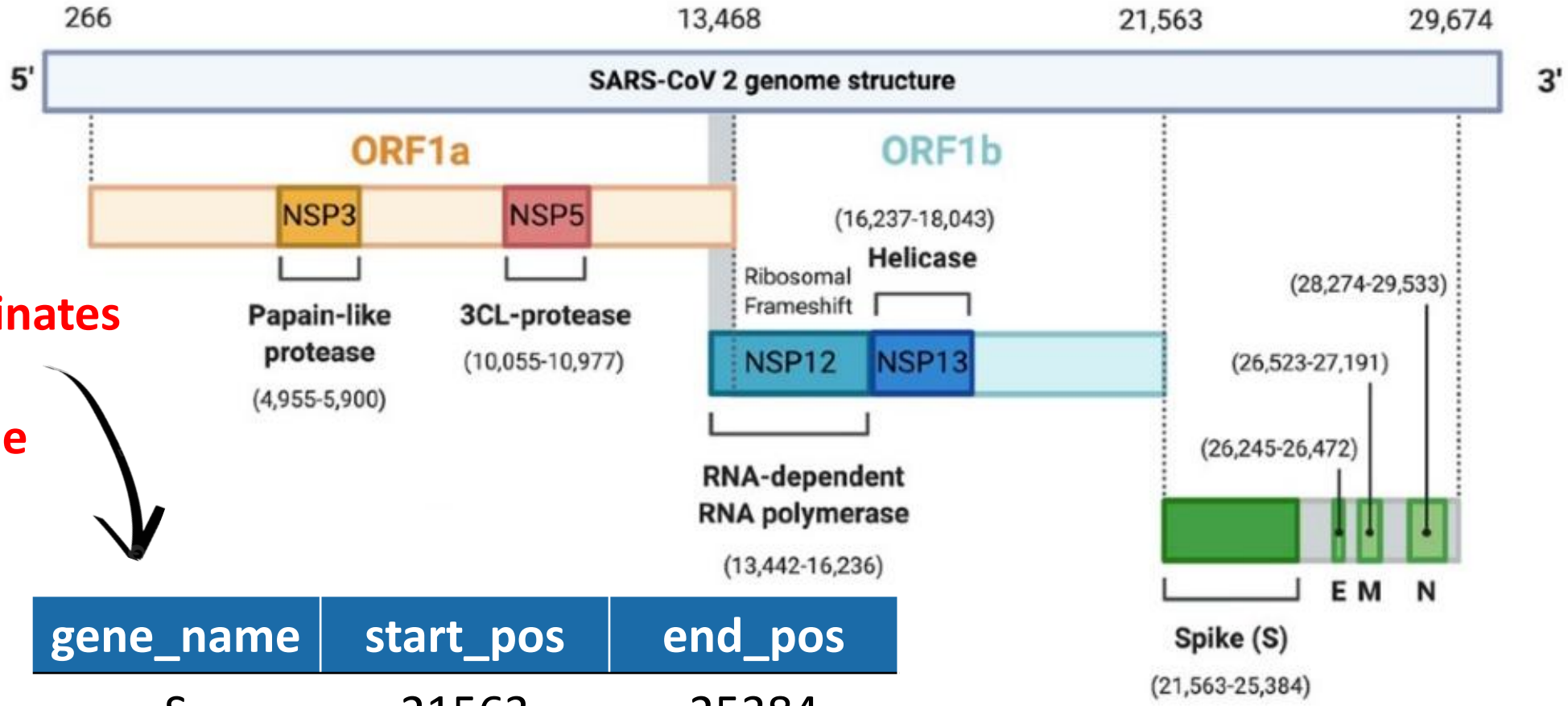
Mutation	Position	Lineage(s)	Significance
K417N	22,813	Beta, Omicron	RBD — reduces antibody neutralisation
E484K	23,012	Beta, Gamma	RBD — immune escape hotspot
N501Y	23,063	Alpha, Beta, Gamma, Omicron	RBD — increased ACE2 binding affinity
D614G	23,403	All modern lineages	Universal — first globally dominant mutation
P681H/R	23,604	Alpha (H), Delta (R)	Furin cleavage site — enhanced cell entry



SARS-CoV 2 virion structure

Structure of SARS-CoV-2 genome (Alanagreh et al., 2020)

SARS-CoV-2 genome information



All variant coordinates are relative to a reference genome

GENE table

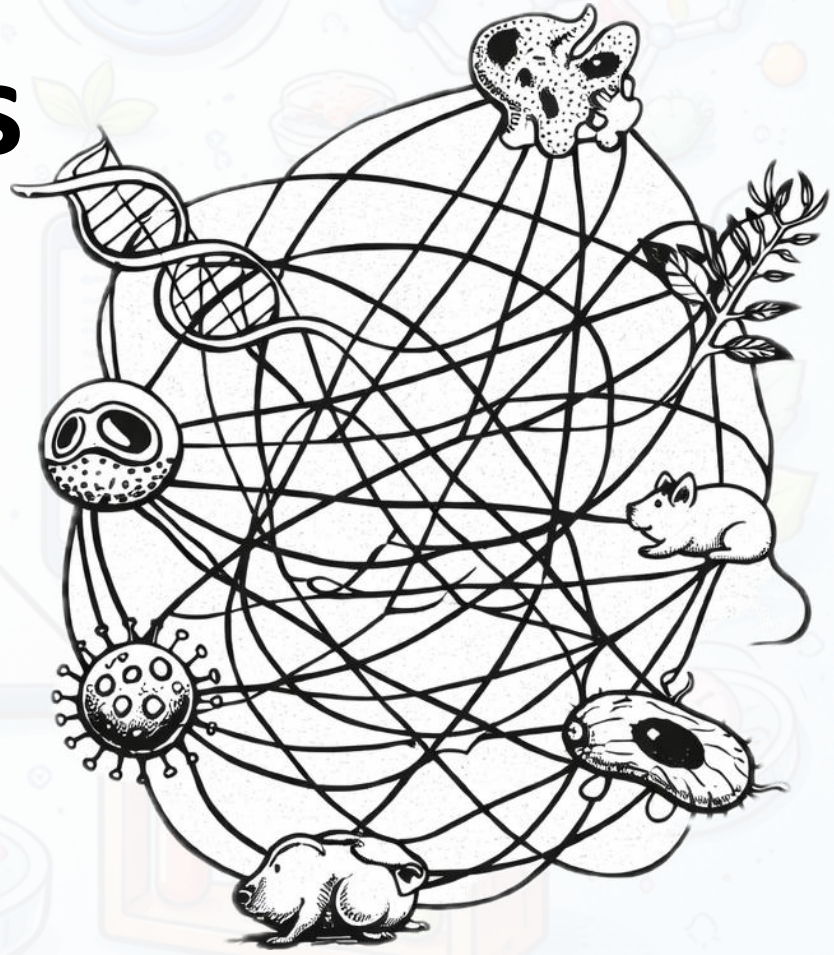
gene_name	start_pos	end_pos
S	21563	25384
ORF1ab	1	21563

Structure of SARS-CoV-2 genome (Alanagreh et al., 2020)

03

VCF Files in Viral Genomics

The lingua franca of variant data



Viral VCF File - anatomy

```
##fileformat=VCFv4.3
##reference=NC_045512.2
##source=iVar v1.3.1
##INFO=<ID=AF,Number=A,Type=Float,Description="Within-sample allele frequency">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total read depth">
#CHROM        POS    ID  REF  ALT  QUAL  FILTER  INFO
NC_045512.2   23063  .   A    T    5.0   PASS    AF=0.997;DP=2341   ← N501Y (fixed in sample)
NC_045512.2   23403  .   A    G    5.0   PASS    AF=1.000;DP=1887   ← D614G (universal)
NC_045512.2   22813  .   G    T    3.1   PASS    AF=0.043;DP=1654   ← K417N (minority variant)
NC_045512.2    6402  .   T    C    1.2   LowQual AF=0.011;DP=234    ← probable error, excluded
```

CHROM

NC_045512.2 — not 'chr22'. For influenza, this would be the segment name (HA, NA…)

AF = 0.997

Near-fixed in this sample — 99.7% of reads support the variant. Viruses are haploid: no 0/1 genotype.

AF = 0.043

Minority variant — ~4% of viral particles carry K417N. Within-host evolution in action.

LowQual

Excluded from analysis. Low depth (234×) + low AF (1.1%) → likely sequencing error.

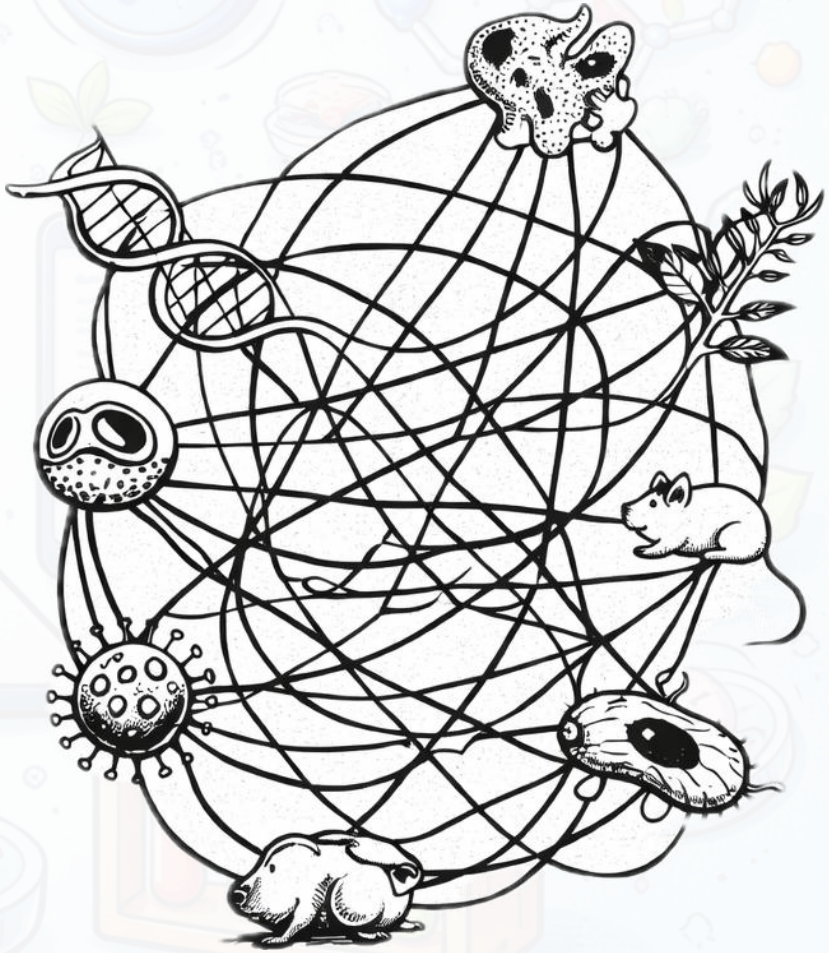
Viral VCF File – allele frequency interpretation

AF = 1.0	**Fixed** Every viral particle in this sample carries the mutation. Defines the lineage.
AF > 0.5	**Major variant** Dominant allele. May be spreading within the host or mixed infection.
0.05–0.5	**Minority variant** Sub-population co-exists. Could be new mutation or residual ancestral virus.
0.01–0.05	**Noise floor** Borderline. Illumina error ~0.1–1%. Use with caution — require DP ≥ 200.
AF < 0.01	**Sequencing error** Exclude. At this level the expected error rate exceeds the signal.

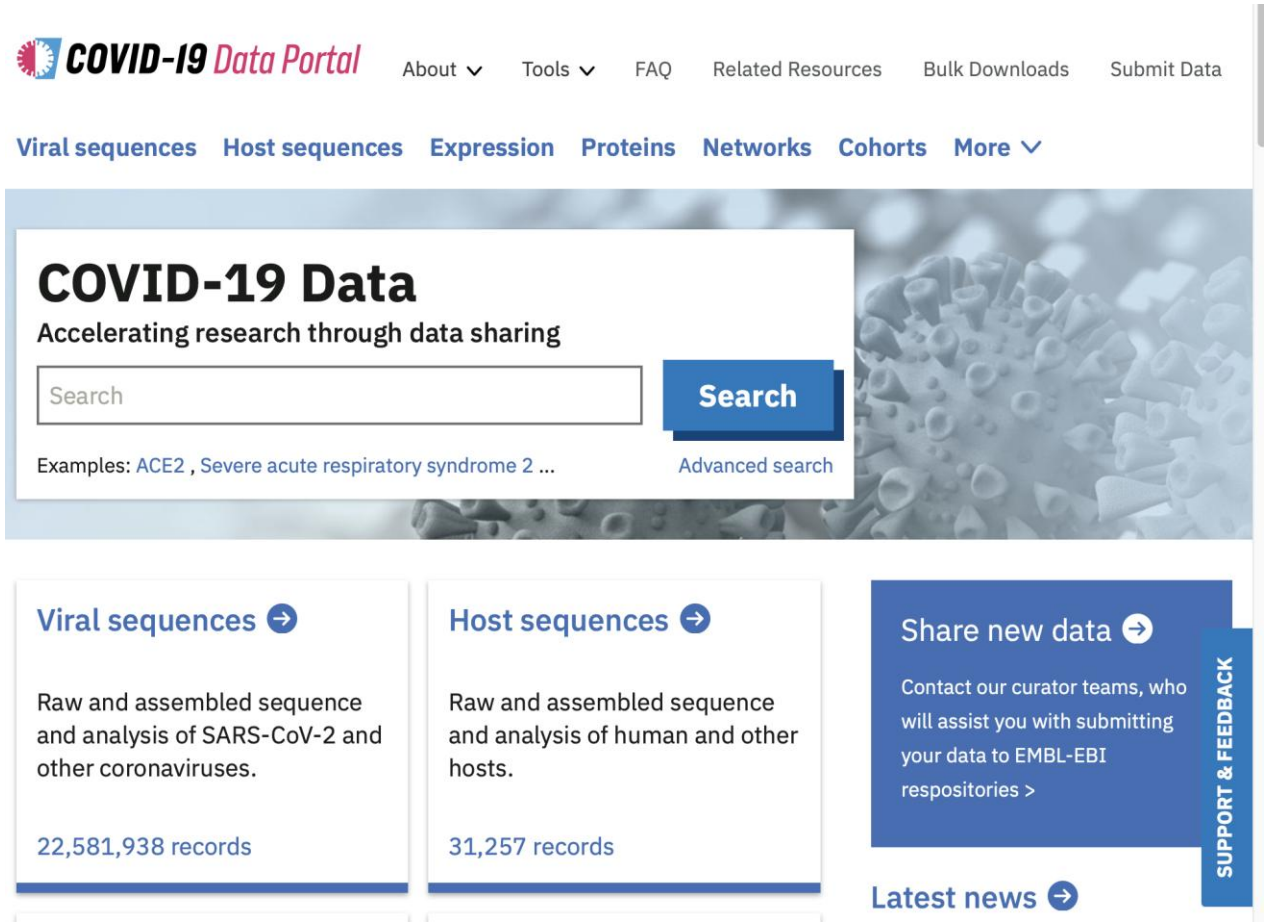
04

Group Studies

CoVEO · ViralPrimer · SewageGPT



CoVEO database



https://k8plex-veo.vo.elte.hu/notebook/report/kpapp-coveopcr/

CoVEO data model - simplified

metadata
runid PK
collection_date
host_id FK
country_id FK
variant_id FK
instrument_id FK

1:N

vcf_key
key PK
runid FK
pos
ref / alt
af
dp

1:1

annotation
key PK
annotation_impact
fene_name
feature_id
hgvs_p
aa_pos

country
id PK
country_name
population_2021

Design: one row per mutation per sample in variants — enables fast GROUP BY across lineages, countries, and time.

Quering CoVEO

Answering real biological questions with SQL

Top 10 most frequent Spike mutations in all Omicron samples

```
SELECT v.pos, v.ref, v.alt, COUNT(*) AS n_samples, AVG(v2.af) AS mean_af
FROM datahub_0.vcf_key v
INNER JOIN datahub_0.vcf v2  ON v.key = v2.key
INNER JOIN datahub_0.lineage l ON v.runid = l.runid
WHERE v.pos BETWEEN 21563 AND 25384
AND l.variant_id = 'Omicron' AND v2.af > 0.5
GROUP BY v.pos, v.ref, v.alt
ORDER BY n_samples DESC
LIMIT 10;
```

	pos	ref	alt	n_samples	mean_af
0	23055	A	G	300820	0.990131
1	22995	C	A	300820	0.992735
2	23040	A	G	300820	0.994651
3	22679	T	C	300820	0.992683
4	22898	G	A	300820	0.988221

A compatible analysis

CoVEO database
1.6+ samples

2021 April

GISAID database
8.3M+ samples

Article | Open access | Published: 04 November 2022

Identification of mutations in SARS-CoV-2 PCR primer regions

Anikó Mentes, Krisztián Papp, Dávid Visontai, József Stéger, VEO Technical Working Group, István Csabai, Anna Medgyes-Horváth & Orsolya Anna Pipek

Scientific Reports **12**, Article number: 18651 (2022) | Cite this article

6754 Accesses | **23** Citations | **6** Altmetric | Metrics

ViralPrimer server

VIRALPRIMER

Home | Search | Browse | Primer Mapping | Mutation Analysis | Help/About | Statistics | Tutorials

Welcome to ViralPrimer

"The Web Server to Monitor Viral Nucleic Acid Amplification Tests' Primer Efficiency during Pandemics"

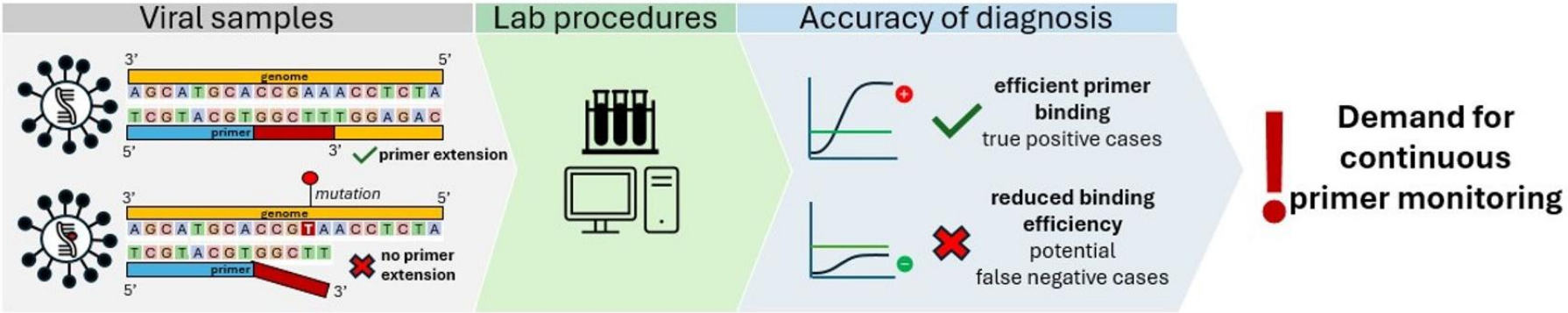
About

We offer a comprehensive list of SARS-CoV-2 genomic mutations especially focusing on mutations affecting 50 primer target regions utilized in diagnostic, sequencing PCR, and LAMP techniques. Our analysis is based on a vast dataset from the European COVID-19 Data Portal (https://www.covid19dataportal.org/) portal (condensed into the CoVEO database), comprising over 2 million samples, and the GISAID database (https://gisaid.org/), which contains more than 15 million SARS-CoV-2 samples. Additionally, we provide a curated list of mutations for Mpox, indicating their impact on PCR efficiency. Our ultimate goal is to facilitate rapid diagnostics for future pandemics by the real-time monitoring of genomic variations influencing primer efficiency.

The ViralPrimer web server can be accessed for free.

Publication: Deutsch et al, 2024 https://doi.org/10.1093/bioinformatics/btae657

ViralPrimer server



A SARS-CoV-2 mutation database with primer impact analysis

19,007,636 samples

GISAID (16.2M) + COVID-19 Data Portal (2.8M) processed through a unified VCF pipeline

1,372 primer oligos

From 50 primer sets: 10 LAMP, 4 sequencing, 35 diagnostic PCR — all mapped to mutation positions

Kooplex platform

CoVEO web app built on Kooplex: direct DB access + Jupyter notebooks for advanced analysis

Lineage tracking

Samples classified into WHO Variants of Concern (VOC) and Variants Under Investigation (VUI)

Publication: Deutsch et al, 2024 https://doi.org/10.1093/bioinformatics/btae657

Sewage database and GPT

Core question: Can a large language model translate plain English into correct SQL for a complex, domain-specific biological database?

239 sewage samples
5 European cities: Budapest, Copenhagen, Rotterdam, Rome, Bologna
GPT-4 vs Claude: Direct comparison of two leading LLMs for Text-to-SQL accuracy

Article | Open access | Published: 22 November 2025

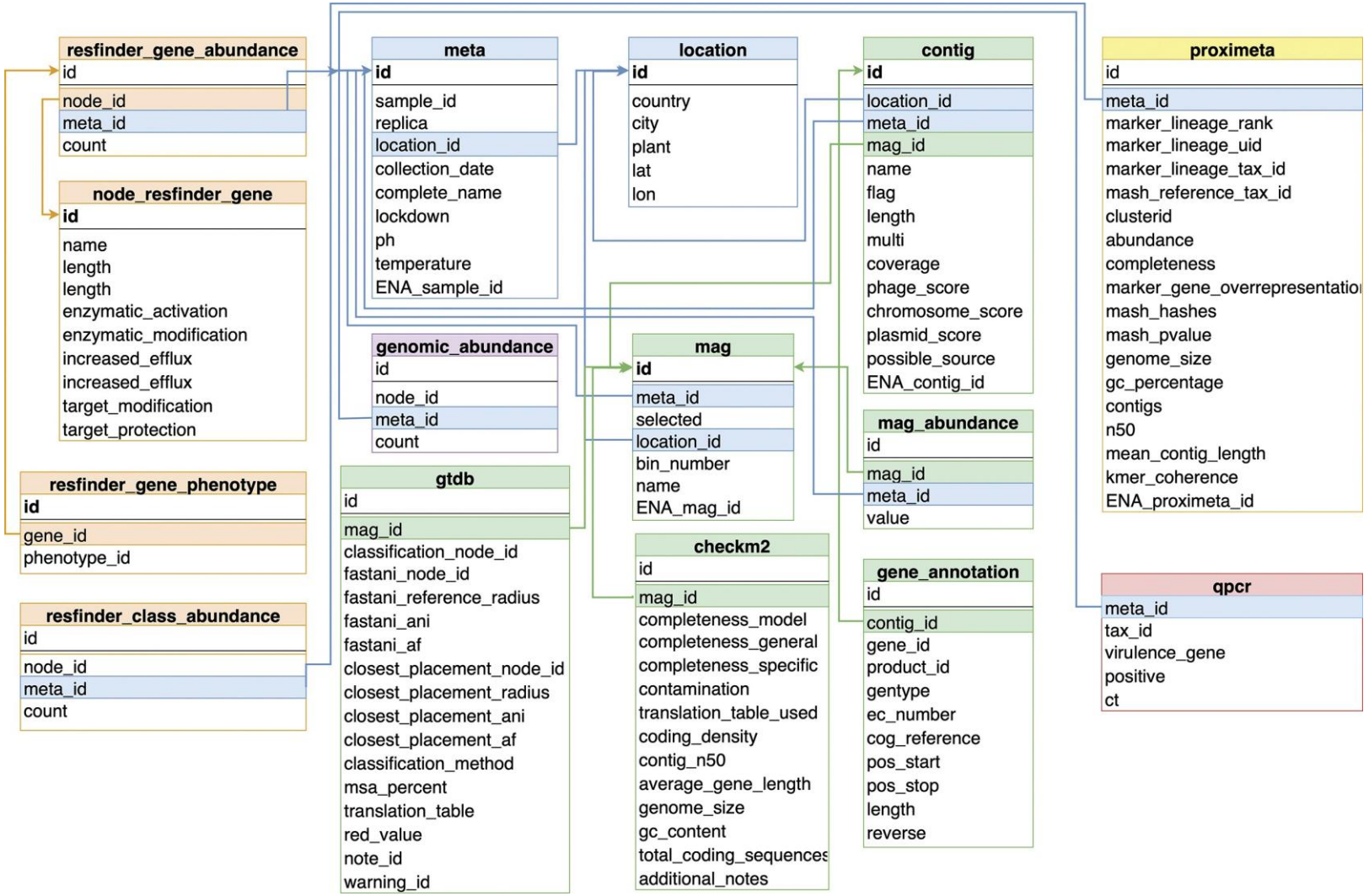
A case study using sewage metagenomic data for assessment of text-to-SQL capabilities in large language models

Ágnes Becsei, József Stéger, Dávid Visontai, Patrick Munk, Frank M. Aarestrup, István Csabai & Krisztián Papp

Scientific Reports **15**, Article number: 44557 (2025) | Cite this article

3397 Accesses | Metrics

Sewage database



Publication: Becsei et al, 2026 https://doi.org/10.1093/database/baaf089

SewageGPT architecture

User Question

Plain English
'Show me ARG trends in Budapest 2022'

LLM (GPT-4 / Claude)

System prompt:
Schema + biology glossary

SQL Query

SELECT … FROM …
JOIN … WHERE …
GROUP BY …

Result + Plot

Table returned, narrated by LLM, visualized in Python

Common failure modes:

Ambiguous question

'Show me resistance genes' → LLM defaults to SELECT * — no filter, no aggregation

Column name clash

'abundance' exists in 3 tables → wrong JOIN without explicit table prefix

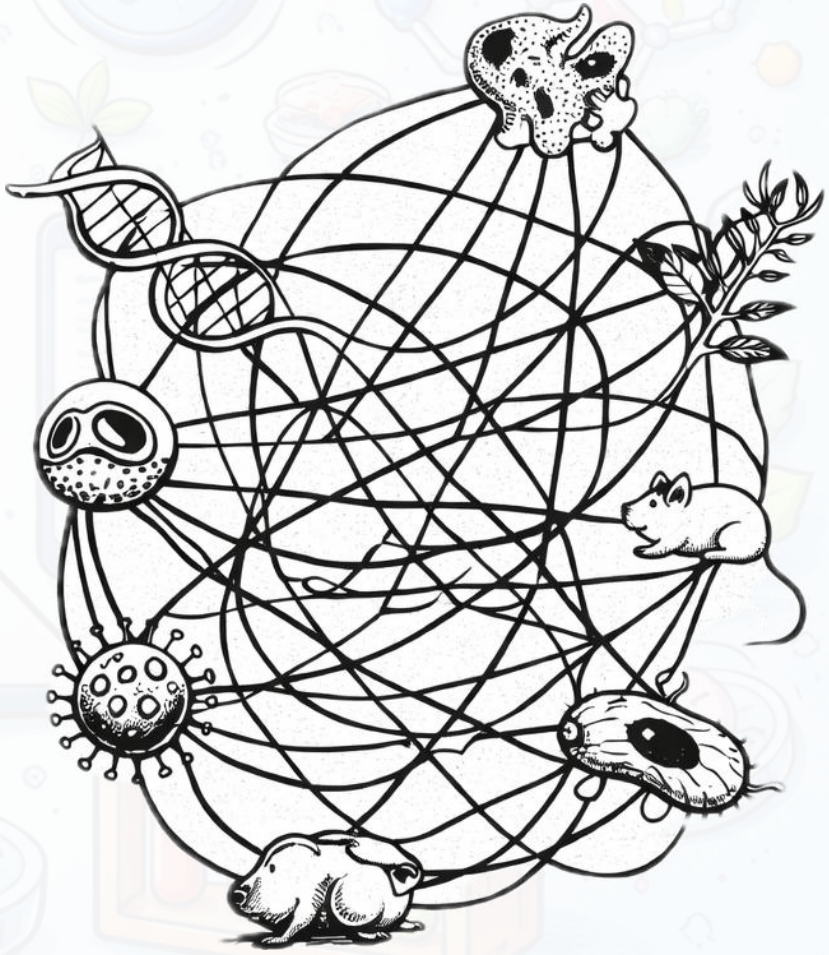
Taxon spelling

Misspelled species name → silent empty result (not an error — just 0 rows returned)

05

AI-Assisted Analysis

GitHub Copilot · VSCode · MCP Servers



VSCode + GitHub Copilot

Essential VSCode extensions

GitHub Copilot
Inline LLM code completion — just write a comment and press Tab

Remote – SSH
Edit code locally, run on HPC cluster or cloud VM

Python + Pylance
Core analysis: pandas, cyvcf2, pysam, biopython

Jupyter Extension
Notebook interface inside VSCode

Rainbow CSV
Colour-codes VCF/TSV columns for manual inspection

GitLens
Git blame and history — essential for reproducibility

Copilot in action

```
# Read VCF, filter PASS variants in Spike
# with AF > 0.8 from Omicron samples
# Return DataFrame with POS, REF, ALT, AF

import cyvcf2, pandas as pd

vcf = cyvcf2.VCF('sample.vcf.gz')
records = []
for v in vcf:
    if (v.FILTER == 'PASS'
            and v.INFO.get('AF', 0) > 0.8
            and 21563 <= v.POS <= 25384):
        records.append({
            'POS': v.POS, 'REF': v.REF,
            'ALT': v.ALT[0],
            'AF': v.INFO['AF']})
df = pd.DataFrame(records)
```

Model Context Protocol (MCP)

MCP: an open standard (Anthropic, 2024) that lets LLMs call external tools — query a database, run a command, read a file, call an API.

User (VSCode or Claude app)

Types natural language request

LLM (Claude / GPT-4)

Decides which tool to call

MCP Client

Routes tool call JSON-RPC / stdio

MCP Servers

filesystem · postgres
shell · Ensembl API
NCBI Entrez

```
# ~/.config/claude/mcp.json — connect Claude Desktop to your databases
{
  "filesystem": { "command": "npx", "args": ["@modelcontextprotocol/server-filesystem", "/data/vcf"] },
  "postgres":   { "command": "npx", "args": ["@modelcontextprotocol/server-postgres",
"postgresql://localhost/sewage_db"] },
  "shell":      { "command": "python", "args": ["mcp_shell_server.py"] }
}
```

AI tools – critical caveats

- **Hallucinated API calls:** LLMs may generate plausible but wrong bioinformatics tool flags. Always test on a small example first.

- **Reference genome confusion:** Specify NC_045512.2 or GRCh38 explicitly. The LLM may silently pick the wrong assembly.

- **MCP trust boundary:** A shell MCP server can execute any command. Run only on trusted machines; audit permissions carefully.

- **Ambiguity cascades:** Vague questions → wrong SQL → wrong result — with no error message. Human review is non-negotiable.

- **Reproducibility:** Log every prompt, model version, and generated SQL. Treat LLM-generated code like any other code: version control it.

Coming up - notebook with KooplexQuery

- **Notebook with Text-to-SQL AI assistent results**

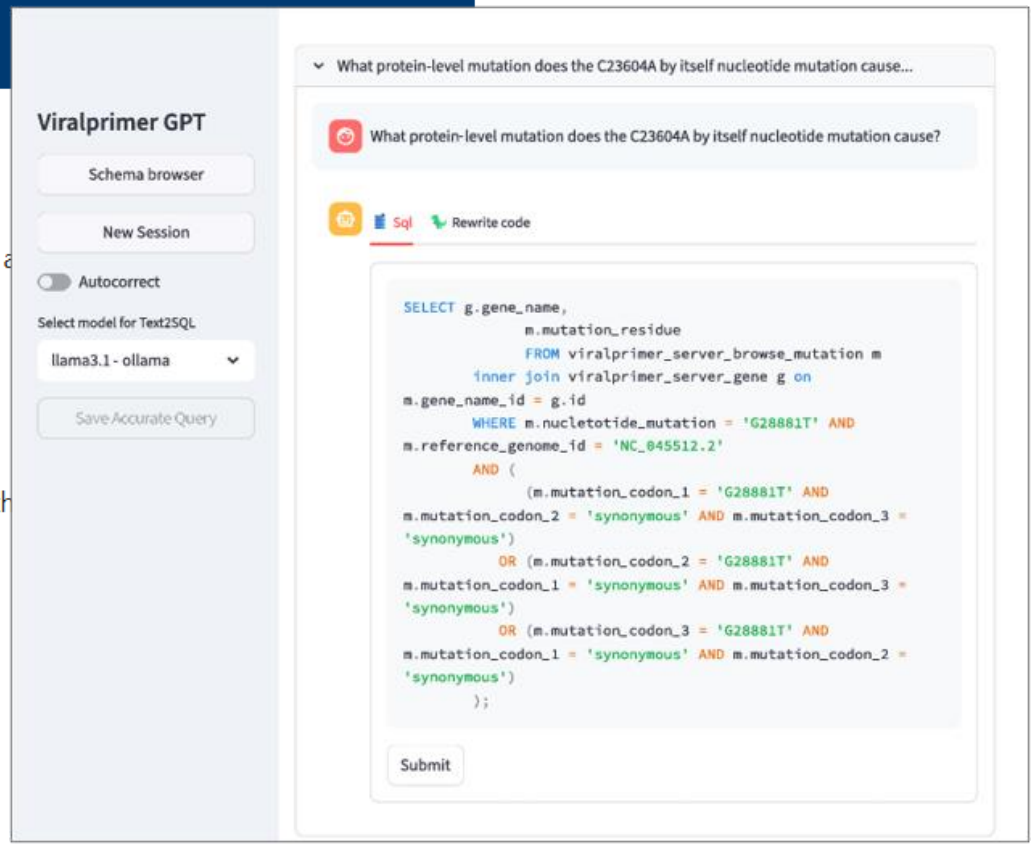
From Scientific Question to SQL: AI-Assisted Viral Genomics

A hands-on class for physics students | ViralPrimer Database | ELTE TTK

What this notebook is about

We will work with a real PostgreSQL database called **ViralPrimer**, which tracks nucleotide mutations in SARS-CoV-2 a
surveillance datasets (GISAID and CoVEO). Along the way, we will:

1. Understand the biological context at just enough depth to write meaningful queries
2. Practice translating scientific questions into SQL — step by step
3. See how **AI coding assistants in VSCode** can help — and where they fall short without context
4. Experience the difference that a **schema-aware AI assistant** (KooplexQuery) makes when the AI has access to th



Summary

CoVEO / ViralPrimer: doi.org/10.1093/bioinformatics/btae657
SewageGPT: doi.org/10.1038/s41598-025-28341-7
ViralPrimer web server: viralprimer.elte.hu
MCP documentation: modelcontextprotocol.io